Data Analytics and Machine Learning in Modern Cricket

Abstract

This paper reviews how data analytics and machine learning are used to quantify player performance and predict match outcomes in modern cricket. We describe the core feature sets used across batting, bowling, and fielding metrics, the rise of win-probability models, and the practical trade-offs between interpretable statistical models and black-box machine learning approaches for team selection and in-match strategy.

1. Introduction

Traditional cricket statistics such as batting average and bowling economy rate summarize a player's career but say little about context -- the quality of opposition, the match situation, or the pitch conditions faced. Modern cricket analytics instead builds context-aware metrics, most notably variants of Expected Runs and Player Impact scores that adjust raw output for the difficulty of the situation in which it was produced.

The growth of ball-by-ball tracking data -- pitch length, line, shot type, field placement -- has made it possible to move from season-level summaries to per-delivery models, which is the granularity most modern analytics teams now operate at.

2. Win Probability Models

A win probability model estimates, at any point in a match, the likelihood that a given team will win, using inputs such as current score, wickets remaining, overs remaining, and the par score for the format. These models are typically trained on historical ball-by-ball data using logistic regression or gradient-boosted trees, and are the basis for the live win-probability graphics shown during broadcasts. In T20 cricket specifically, win probability swings much faster than in ODIs or Test cricket because a single over can shift the required run rate dramatically.

3. Player Performance Prediction

Predicting a player's performance in an upcoming match combines recent form, historical performance against similar opposition bowling or batting styles, and venue-specific tendencies. Random forest and gradient-boosting models are common choices because they handle the mixed categorical and numeric features involved (player identity, opposition, venue, format) without heavy manual feature engineering, while still allowing feature-importance analysis to explain predictions to non-technical stakeholders such as selectors.

4. Limitations

Machine learning models trained on historical data can struggle to adapt quickly to genuine changes in a player's game -- a bowler who has just added a new variation, for example -- since the model has not yet seen enough data reflecting the change. Analytics teams typically combine model output with expert coaching judgment rather than relying on the model as a sole decision-maker, particularly for high-stakes selection decisions.